

Task Title: Chat with Page Feature Integration

GitHub Repository Link

GitHub Repository:

<https://github.com/SANMOLDEEP/vikaspedia-internship/tree/main/task2/chatpage>

Description

This task focuses on the design and implementation of an interactive AI-powered “Chat with Page” feature integrated into a ReactJS application. The main objective of this task is to build a contextual chatbot that answers user queries strictly based on the content available on the current webpage.

The application is developed using ReactJS with standard CSS for styling. A floating chat button is positioned at the bottom-right corner of the screen, placed above the existing Annotation toggle button to avoid overlap. When the user clicks this button, a sidebar smoothly slides in from the right side of the screen. The sidebar contains a header section, a scrollable message area, and an input field with a send button at the bottom. On desktop devices, the sidebar width is approximately 350–400px, while on mobile devices it occupies the full width of the screen to ensure responsiveness.

For AI integration, the Google Gemini API (Free Tier) is used. The API was selected because it offers a reliable free-tier, easy REST integration, good contextual understanding, and proper documentation. Prompt engineering is carefully implemented to ensure that the AI answers strictly using the provided page content. The system instruction clearly states that if the answer is not available in the given context, the AI must respond accordingly instead of generating unrelated information.

The application also includes proper loading states and error handling. When a user submits a query, a “Thinking...” indicator is displayed until the response is received. If the API call fails due to network errors or rate limits, a user-friendly message such as “I’m having trouble connecting right now. Please try again.” is displayed inside the chat window.

This task helped in gaining practical experience with:

- React component architecture and state management
- Building responsive sidebar UI components
- Dynamic DOM content extraction and cleanup
- AI API integration and prompt engineering
- Error handling and loading state management
- Secure API key usage (without hardcoding keys)

Overall, the Chat with Page feature successfully demonstrates how AI can be integrated into a web application to provide intelligent, document-specific assistance, enhancing user interaction and improving the overall usability of content-driven platforms.